Mahima Sachan

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Software engineer with 5+ years at Microsoft & Salesforce, building and scaling enterprise systems serving 300M+ users. Experienced in full-stack development, product engineering, and developing LLM-powered and agentic AI systems for real-world applications.

EXPERIENCE

Microsoft

Technical Product Manager (Engineering-Focused) — Microsoft Purview, Data Lifecycle Management Dec 2023 – Jan 2025

- Led support strategy and operations for DLM, serving 317M users; managed 5k+ monthly tickets and reduced escalations by 22%.
- Defined and launched the Deeply Protected Users (DPU) metric, integrating it into executive dashboards to improve product prioritization for DLM investments.
- Shipped Microsoft Graph APIs for Labels and File Plan Descriptors, enabling third-party integrations with DLM and driving 5M API calls per month.

Software Engineer II — Meeting Insights Mar 2022 – Dec 2023

- Reduced service memory footprint by 32% by analyzing heap dumps to identify object retention and allocation inefficiencies; redesigned core data structures using memory pooling and streaming.
- Built and scaled data ingestion pipelines for emails, documents, meetings, and user signals; integrated into a RAG-based system powering Microsoft Meeting Insights, supporting 300M+ daily active users.

Software Engineer — Windows 11 Voice Access Dec 2020 – Feb 2022

- Led end-to-end development of the Voice Access first-run experience, deployed across 1B+ devices; built a responsive UI framework supporting accessibility features including screen readers, keyboard navigation, high-contrast themes, and multi-resolution layouts.
- Developed voice processing pipelines integrating speech-to-text with intent recognition and contextual app-state understanding, enabling accurate action execution across Windows applications.

Salesforce

Member of Technical Staff — Marketing Cloud Jun 2019 – Dec 2020

- Built REST APIs to fully automate patching and decommissioning across a ~10k CentOS 6 Linux estate, reducing vendor-driven operations from 6 days per server to seconds.
- Designed and developed a full-stack one-click orchestration UI in React, surfacing backend automation to non-technical ops teams and enabling self-serve infrastructure management at scale.

PROJECTS

Improving Reliability of LLMs in Software Engineering *(Ongoing M.S. Thesis)*

- Designing evaluation frameworks for LLM-based software engineering systems using benchmarks such as HumanEval and SWE-bench; analyzing failure modes and improving robustness in code generation and task execution using metrics including pass@k, execution accuracy, and task success rate.

Memory-Centric Architectures for LLM Acceleration

- Extended llama.cpp with a 4-tier disaggregated memory simulator (SRAM/HBM/DRAM/CXL) to empirically prove LLM decode is memory-bound; quantified a 40% throughput drop at scale and showed CXL disaggregation imposes a 40× latency penalty vs. HBM.

Customer Review Sentiment & Purchase Behavior Analytics

- Designed a framework to analyze text-rating inconsistency (TRI) and its impact on return probability, engagement, and revenue outcomes; applied regression models to identify statistically significant relationships ($p<0.05$), improving predictive performance by 15–20% over baseline.

SKILLS

Languages: Python, C++, Java, JavaScript, SQL, C#, Go, Bash, CUDA
Backend & Frameworks: Spring Boot, REST APIs, .NET, ASP.NET Core, FastAPI, Flask, Django, Node.js, React, Next.js, Nginx
AI/ML Systems: LLMs, RAG, MCP, LoRA, LangChain, LangGraph, Vector search, Embeddings, PyTorch, TensorFlow, Scikit-learn
Distributed Systems & Data: Kafka, Spark, ETL pipelines, Elasticsearch
Cloud & DevOps: AWS, Azure, Docker, Kubernetes, CI/CD (Azure DevOps), Git
Databases: SQL Server, MySQL, NoSQL, DynamoDB, S3, Redis, MongoDB
Testing & Quality: Test Driven Development, PyTest, JUnit, NUnit
Security: OAuth2, RBAC, Data governance, Compliance systems

EDUCATION

California State University, Long Beach Jan 2025 – Dec 2026
M.S., Computer Science — GPA: 4.00/4.00 Long Beach, CA
Coursework: Pattern Recognition, Machine Vision, Advanced Algorithms, Advanced AI, Computer Simulation & Modeling, Thesis on improving reliability of LLMs in Software Engineering

Motilal Nehru National Institute of Technology Jul 2015 – May 2019
B.Tech., Computer Science and Engineering — GPA: 3.89/4.00 Prayagraj, India
Coursework: Data Structures & Algorithms, Operating Systems, Database Systems, Distributed Systems, Object-Oriented Modeling